

Experiment 4:

Implement a finite-state machine for morphological parsing. In this example, we'll create a simple machine to generate plural forms of English nouns using python.

Program:

```
word = input("Enter a singular noun: ")

if word.endswith("y"):

    plural = word[:-1] + "ies"

elif word.endswith(("s", "x", "z", "ch", "sh")):

    plural = word + "es"

else:

    plural = word + "s"

print("Plural Form:", plural)
```

Output:

```
word = input("Enter a singular noun: ")

if word.endswith("y"):
    plural = word[:-1] + "ies"
elif word.endswith(("s", "x", "z", "ch", "sh")):
    plural = word + "es"
else:
    plural = word + "s"

print("Plural Form:", plural)
```

Enter a singular noun: city
Plural Form: cities

